

Design and Implementation of a Database Management System for Laboratory Equipment Booking and Maintenance System

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WHY THIS SYSTEM IS NEEDED

Problem Statement

Laboratories currently manage equipment, users, bookings, usage, calibration, consumables and maintenance manually (registers / spreadsheets). This creates conflicts and leaves weak, unreliable equipment histories — with no single source of truth for who used what, when, and in what condition.

1

Booking Conflicts

Manual scheduling leads to double-bookings and clashing reservations for the same equipment / time slot.

2

No Usage History

Equipment usage is not logged consistently, so utilization and downtime cannot be analysed.

3

Calibration Risk

Calibration due-dates are tracked informally, risking non-compliance and inaccurate readings.

4

Consumable Shortages

Stock levels for consumables are not monitored, causing last-minute shortages.

5

Poor Breakdown Records

Breakdowns and vendor servicing are not systematically recorded or followed up.

Project Scope

In Scope

- Conceptual design (ER model) → normalized (3NF) relational schema
- Equipment, booking, usage-log, consumable, calibration, breakdown and maintenance data
- Booking / approval workflow with conflict prevention
- Consumable stock tracking with reorder limits
- Calibration and maintenance due-date tracking
- Java / Python front-end for CRUD operations, search and reporting
- Role-based access for different user types

Out of Scope

- Direct IoT / sensor integration with lab instruments
- Billing, invoicing and financial / payment processing
- Automated third-party procurement or e-commerce integration
- Mobile native application (web / desktop client only)
- Enterprise-wide identity management (institution SSO)

03 Objectives

1

Design a normalized (3NF) relational schema for the laboratory equipment booking & maintenance domain

2

Enforce business rules such as no overlapping approved bookings for the same equipment

3

Maintain an accurate, queryable history of equipment usage, calibration and breakdowns

4

Track consumable stock levels and flag reorder needs automatically

5

Develop a Java / Python front-end supporting CRUD, search, approval and reporting

6

Support role-based access for technicians, managers, maintenance staff and administrators

Users & Roles

U1

Lab Technician / Equipment User

Searches equipment availability, creates bookings, logs actual usage.

U2

Lab Manager / Admin

Approves or rejects bookings, manages equipment, categories and vendors.

U3

Maintenance Staff / Coordinator

Reports breakdowns, schedules and tracks maintenance jobs with vendors.

U4

System Administrator

Manages user accounts, roles and master data across laboratories.

Functional Requirements

Booking & Usage

- Search equipment by category, laboratory and time-slot availability
- Create, approve / reject bookings with status tracking
- Prevent overlapping approved bookings for the same equipment
- Log actual usage start / end time and remarks per booking

Maintenance & Reporting

- Issue consumables and track stock against reorder levels
- Schedule calibration and monitor due dates per equipment
- Log breakdowns and assign maintenance jobs to vendors
- Generate reports: utilization, downtime, calibration-due, vendor service history

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ENFORCED THROUGH THE SCHEMA DESIGN

Key Business Rules & Constraints



No Overlapping Bookings

Only one approved booking per equipment for a given time slot.



Active Equipment Only

Equipment must be in “Active” status to be reservable.



Valid Calibration Dates

Next-due date must be later than the calibration date.



Stock Limits

Consumable stock cannot silently fall below its defined reorder level.



Valid Completion Dates

Maintenance job completion date must be on/after the job date.



Referential Integrity

PRIMARY KEY / FOREIGN KEY, UNIQUE, NOT NULL, CHECK and DEFAULT constraints throughout.

Conceptual Design — Principal Entities

Laboratory

Physical lab housing equipment

EquipmentCategory

Classification of equipment

Equipment

Individual bookable item

User

Technician, manager or admin

TimeSlot

Bookable date / time window

Booking

Reservation of equipment

UsageLog

Actual usage record

Consumable

Supplies used with equipment

Calibration

Calibration event & due date

Breakdown

Reported equipment fault

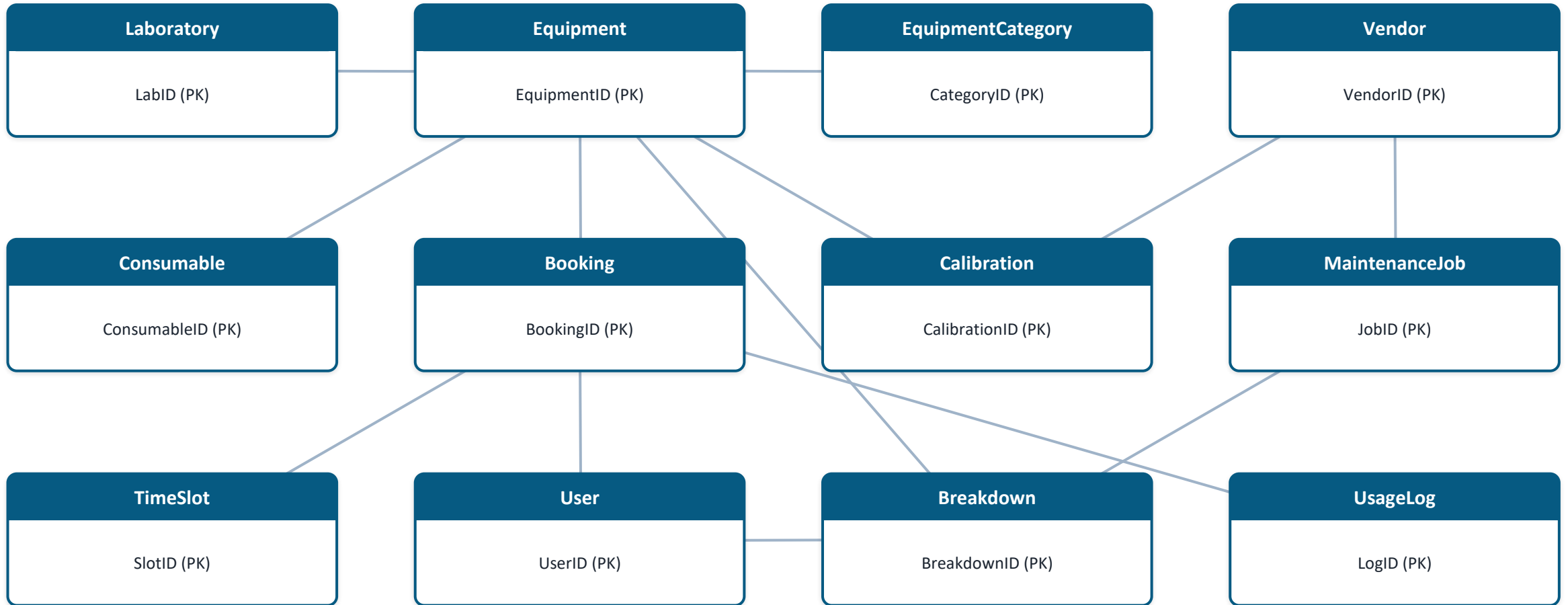
MaintenanceJob

Repair / service job

Vendor

External service provider

Entity-Relationship (ER) Diagram



Initial Relational Schema (1/2)

Table	Primary Key	Foreign Keys	Key Attributes
Laboratory	LabID	—	Name, Location, Capacity
EquipmentCategory	CategoryID	—	CategoryName, Description
Equipment	EquipmentID	LabID, CategoryID	Name, Model, Status
User	UserID	—	Name, Role, Email, Department
TimeSlot	SlotID	—	Date, StartTime, EndTime
Vendor	VendorID	—	Name, ContactInfo, ServiceType

Initial Relational Schema (2/2)

Table	Primary Key	Foreign Keys	Key Attributes
Booking	BookingID	EquipmentID, UserID, SlotID	BookingDate, Status, ApprovedBy
UsageLog	LogID	BookingID	ActualStart, ActualEnd, Remarks
Consumable	ConsumableID	EquipmentID	Name, StockQty, ReorderLevel
Calibration	CalibrationID	EquipmentID, VendorID	CalibrationDate, NextDueDate
Breakdown	BreakdownID	EquipmentID, ReportedBy (UserID)	ReportedDate, Description, Severity
MaintenanceJob	JobID	BreakdownID, EquipmentID, VendorID	JobDate, CompletionDate, Cost, Status

Schema Relationships Overview

Laboratory (1) — (N) Equipment

EquipmentCategory (1) — (N) Equipment

Equipment (1) — (N) Booking

Equipment (1) — (N) Consumable

Equipment (1) — (N) Calibration

Equipment (1) — (N) Breakdown

User (1) — (N) Booking

TimeSlot (1) — (N) Booking

Booking (1) — (N) UsageLog

Vendor (1) — (N) Calibration

Vendor (1) — (N) MaintenanceJob

Breakdown (1) — (N) MaintenanceJob

User (1) — (N) Breakdown (reported by)



Thank You!